CHETAN BORSE

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EDUCATION

M.S., Robotics and Autonomous Systems Graduating May 2026
Arizona State University, Tempe, AZ 4.0 GPA
Ira Fulton School of Engineering
Relevant coursework: Modelling and Control of Robots, Multi-Robot Systems, Perception in Robotics, Reinforcement Learning

TECHNICAL SKILLS

Programming: Python, MATLAB, Arduino, C++, Bash
Robotics: ROS, GazeboSim, MuJoCo, NVIDIA Warp, MPC, PyTorch
Design/Manufacturing: 3D modeling and Drafting (Solidworks, Fusion360), Cura, 3D Printing
Soft Skills: Collaboration & Teamwork, Communication & Documentation, Adaptability & Initiative

RESEARCH EXPERIENCE

IRIS Lab, Tempe, AZ: Master's Thesis Student August 2025 – Present

- Developed *ComFree-Sim*: A GPU-Parallelized Analytical Contact Physics Engine for Scalable Contact-Rich Simulation and Control, submitted to IROS 2026 (under review).
- Achieved up to 5$\times$ improvement in simulation step speed through GPU-parallelization, enabling large-scale contact-rich simulation.
- Developed sampling-based Model Predictive Control (MPPI) for dexterous manipulation, enabling efficient trajectory optimization and real-time deployment ($\sim$20 Hz) on hardware.

PROFESSIONAL EXPERIENCE

Lycan Aerospace, Bangalore, India: Design Engineer April 2023 – May 2024

- Designed structural and payload components for AGROS, a 10L-capacity pesticide-sprayer quadcopter, including detailed 2D manufacturing drawings.
- Collaborated with systems integration and embedded teams while coordinating with suppliers to ensure design intent and manufacturability.
- Utilized in-house 3D printing for rapid prototyping, validation, and iterative refinement of structural components.

Tata Consultancy Services, Mumbai, India: Systems Engineer October 2021 – May 2022

- Supported migration from legacy mainframe systems to an AWS-based Java environment for a U.S.-based client.
- Monitored batch processes, generated reconciliation reports, and resolved data discrepancies.

PROJECTS

Sampling-Based MPC for Manipulation using ComFree-Sim December 2025 - February 2026

- Built a sampling-based MPC pipeline for robot manipulation tasks using a GPU-parallelized contact physics engine.
- Leveraged parallel simulation for faster trajectory optimization in contact-rich environments.
- Demonstrated efficient and scalable control for dexterous manipulation tasks.

Path Planning for Autonomous Underwater Vehicle (AUV) March 2025 - May 2025

- Simulated AUV navigation in GPS-denied environments using ROS Gazebo (UUV Simulator).
- Implemented A* planning with LiDAR-based obstacle detection and occupancy grid mapping.
- Fused DVL and IMU data for odometry estimation.

Debris Detection using Swarm Robots October 2024 - December 2024

- Designed a decentralized MATLAB-based mapping algorithm for multi-agent debris localization for surface vessels.
- Implemented distributed map-sharing to merge local maps into a unified global debris field map.